

A Source-Iteration Framework for Monte Carlo Helmholtz Solvers via Walk-on-Boundary

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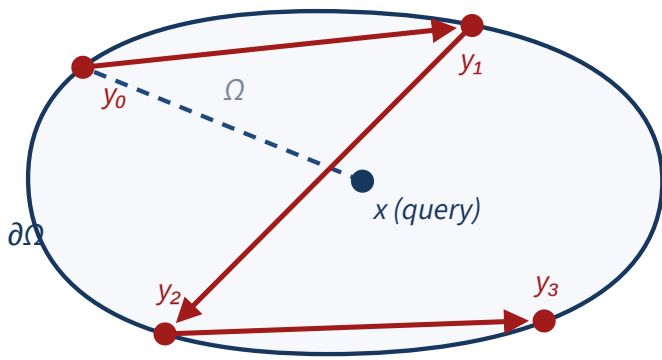
The problem

Simulating time-harmonic waves — sound, light, seismic — means solving $(\Delta + k^2)u = -f$. At the frequencies that matter, every standard route is expensive, and the grid-free Monte Carlo route does not converge at all.

- **Grid solvers pay a pollution penalty.** DOFs must grow faster than k^d , and adjoint / FWI inversion holds $O(k^d)$ wavefield memory per frequency.
- **Monte Carlo sidesteps both:** grid-free and output-sensitive — evaluate u only where asked; no mesh, no global system, no stored field [1]. But the operator is sign-indefinite, so its convergence theory fails.

What is Walk-on-Boundary?

A random walk that lives on the boundary: each hop samples a new boundary point from the Green's-function kernel, and the walk's weighted contributions estimate the boundary density μ .



- Each hop $y_i \rightarrow y_{i+1}$ is one application of the boundary operator A — a single unbiased sample, no mesh anywhere.
- Classic WoB chains many hops to estimate A^n ; my method takes **one hop per iteration** instead.

Why it breaks for Helmholtz

$$(I - A)\mu = b, \quad \mu = \sum_{n \geq 0} A^n b$$

converges only if $\rho(A) < 1$

- At moderate to high k the indefinite kernel gives $\rho(A) \geq 1$: the series **diverges**, and deep walks estimating A^n have **unbounded variance** — the failure the Convergent Born Series [3] first exposed.

Related Work

- **Grid-free MC solvers.** WoS for (screened) Poisson [1]; Walk-on-Stars adds Neumann and Robin data.
- **Walk-on-Boundary.** Practical boundary-integral estimator for Dirichlet / Neumann / Robin data [2]; convergence guarantees are missing for indefinite operators.
- **Convergent Born Series.** Complex wavenumber $k_0^2 + i\varepsilon$ and $\gamma = (i/\varepsilon)V$ force $\rho(M) < 1$ unconditionally [3] — the deterministic precedent for my shift.
- **Closest precedent.** WoS for Helmholtz / Yukawa via the Duffin correspondence [4] — but tied to constant potentials.

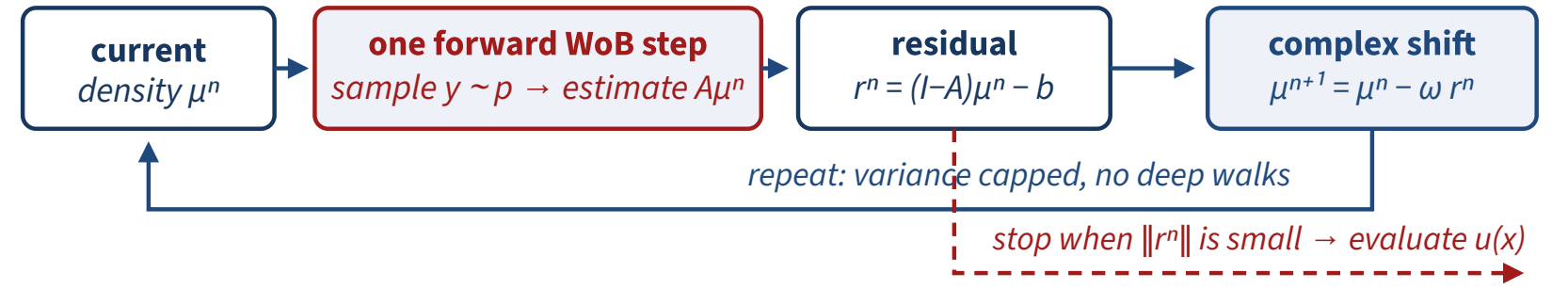
References

- [1] R. Sawhney, K. Crane. 2020. *Monte Carlo Geometry Processing: A Grid-Free Approach to PDE-Based Methods on Volumetric Domains*. ACM TOG 39(4), Art. 123.
- [2] R. Sugimoto, T. Chen, Y. Jiang, C. Batty, T. Hachisuka. 2023. *A Practical Walk-on-Boundary Method for Boundary Value Problems*. ACM TOG 42(4), Art. 81.
- [3] G. Osnabrugge, S. Leedumrongwatthanakun, I. M. Vellekoop. 2016. *A Convergent Born Series for the Inhomogeneous Helmholtz Equation in Arbitrarily Large Media*. J. Comput. Phys. 322, 113–124.

Methodology

The source iteration, one turn of the loop

one iteration = one operator apply



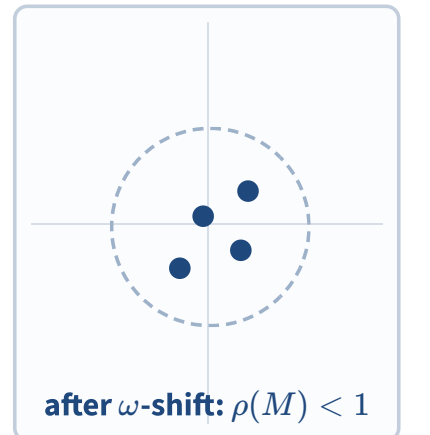
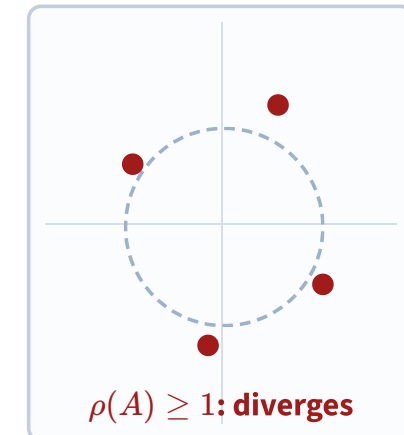
$$\mu^{n+1} = \mu^n - \omega [(I - A)\mu^n - b]$$

$$M = I - \omega(I - A), \quad \rho(M) < 1$$

$$\widehat{A\mu^n} = \frac{K(x,y)\mu^n(y)}{p(y)}, \quad y \sim p$$

One boundary sample per apply: unbiased, matrix-free, variance independent of n .

- **No deep walks.** Every operator apply is one forward WoB step, so per-iteration variance is capped — the exponential blow-up of estimating A^n disappears.
- **Warm starts.** Bootstrap μ^0 from a geometrical-optics / Kirchhoff approximation, a neighbouring frequency, or a learned cache $\hat{\mu}_\theta$ trained by a semi-gradient residual loss.
- **Implementation.** Matrix-free applies and forward walks on **Dr.Jit** [5] with Mitsuba-style ray tracing: JIT-compiled, GPU/CPU-retargetable, differentiable.



A complex ω rotates and rescales the spectrum of $I - A$ into the unit disk about 1 — the stochastic analog of the CBS ε -shift [3].

Evaluation

What I will trust as evidence — figures and table are placeholders, not measurements.

Figure 1 — residual vs. iteration (placeholder)

Figure 2 — $\rho(M)$ over the complex ω plane (placeholder)

Figure 1. $\log \|(I - A)\mu^n - b\|$ vs. iteration n , source iteration against the Neumann series over $\rho(A) \in \{0.5, \dots, 1.5\}$. Falsifiable prediction: Neumann curves turn upward at $\rho \geq 1$; source iteration descends at every ρ .

Figure 2. Measured decay rate as a function of $\{\text{Re } \omega, \text{Im } \omega\}$. Prediction: a convergent basin bounded by the contour $\rho(M) = 1$, with an optimum ω^* inside it.

Table 1 (illustrative placeholder). Outcome vs. spectral-radius sweep — the crossover at $\rho = 1$ is the test.

Method	$\rho = 0.7$	$\rho = 0.95$	$\rho = 1.05$	$\rho = 1.3$
Neumann-series WoB	converges (slow)	converges (very slow)	diverges	diverges
Source iteration (tuned ω)	converges	converges	converges	converges

- **Reference solutions.** Analytic fields where they exist, otherwise a dense deterministic boundary-element solve — error reported against both, never against a single baseline.
- **Controls.** Sweep $\rho(A)$ across the $\rho = 1$ boundary; report per-iteration variance and sample counts, so convergence claims cannot hide behind extra samples.

What it means

1. **Convergence becomes a design choice, not a property of the domain.** One complex scalar ω buys $\rho(M) < 1$ where the Neumann series has none.
2. **Robustness is paid for in rate.** Following the CBS law $\Delta r = 2k_0/\varepsilon$, a larger shift guarantees convergence but slows each iteration — warm starts buy the rate back near $\rho \approx 1$.

[4] X. Yang, A. Rasila, T. Sottinen. 2017. *Walk On Spheres Algorithm for Helmholtz and Yukawa Equations via Duffin Correspondence*. Methodol. Comput. Appl. Probab. 19(2), 589–602.

[5] W. Jakob, S. Speierer, N. Roussel, D. Vicini. 2022. *Dr.Jit: A Just-In-Time Compiler for Differentiable Rendering*. ACM TOG 41(4), Art. 124.

3. **Open questions I want to discuss.** How far does a scalar ω stretch before an operator-valued shift is needed for heterogeneous media (Lippmann–Schwinger), and can $\hat{\mu}_\theta$ warm-start inverse problems without biasing gradients?